Worksheet: Profit and the Price of Cheerios

ECON 201

Work with a neighbor. Write your answers in the spaces.

Three definitions you will use throughout this module. *Total revenue* is the number of units sold times the price per unit: $R = P \times Q$. *Total cost*, written $C(Q)$, is the cost of producing Q units. *Profit* is total revenue minus total cost:

$$\text{Profit} = \underbrace{P \times Q}_{\text{revenue, } R} - \underbrace{C(Q)}_{\text{total cost}}$$

1. Do higher prices always mean higher profits?

A firm's profit is $P \times Q - C(Q)$. To maximize profit, why not just charge a really high price? Jot down your thoughts, then compare answers with a neighbor.

2. Pricing Apple Cinnamon Cheerios

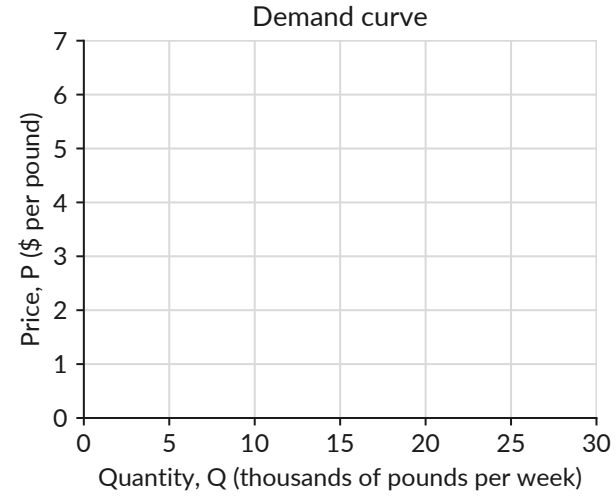
You are the manager at General Mills who sets the price of Apple Cinnamon Cheerios. Producing the cereal costs $2.00 per pound, so producing Q pounds costs $C(Q) = 2Q$. Market research, based on the demand curve the economist Jerry Hausman estimated for this cereal, tells you how many pounds a typical city buys each week at each price:

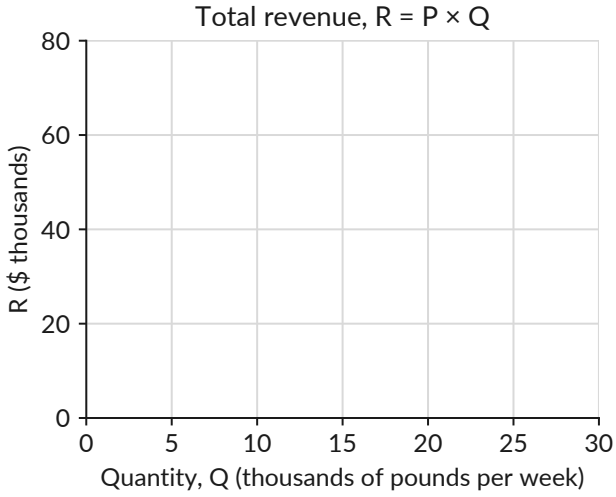
Price per pound, P	$3.00	$3.50	$4.00	$4.50	$5.00
Pounds sold per week, Q	25,000	21,000	17,000	13,000	9,000
Total revenue, R					
Total cost, C					
Profit					

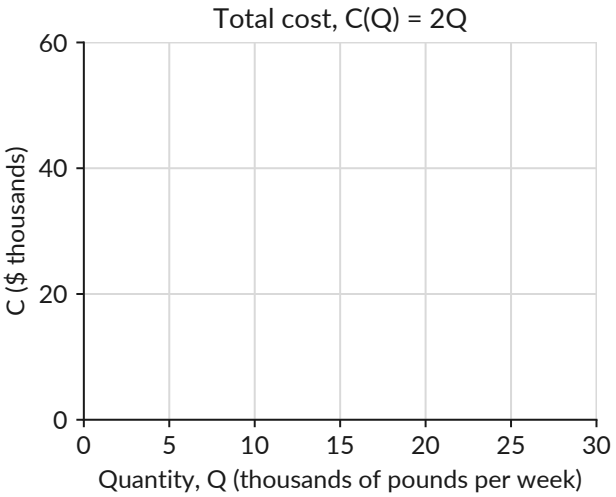
(a) Fill in total revenue, total cost, and profit at each price.

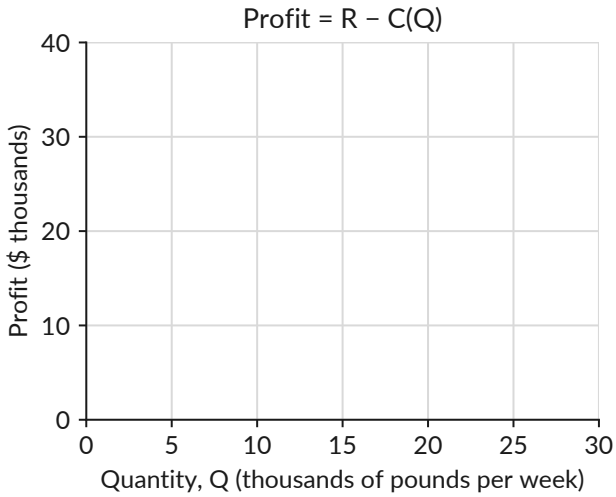
(b) Which price would you set, and what is the weekly profit there?

(c) Plot the five columns of your table on the four grids below, one point per column on each grid, and connect the points: the demand curve (P against Q), total revenue R, total cost C, and profit.









(d) Total revenue is highest at \$3.00, yet that is not the most profitable price. Explain why in a sentence or two.

(e) Why does profit fall when you raise the price from \$4.00 to \$5.00, even though you earn a bigger margin on every pound?